

OpenClaw

AI-Powered Project Operations & Document Intelligence

OpenClaw is an AI-assisted project operations and document intelligence platform for engineering teams.

Built with **FastAPI + React/Vite** and powered by **Mistral AI**, OpenClaw demonstrates how AI, continuous monitoring, document intelligence, and workflow automation can reduce repetitive project-management activities and surface issues requiring human attention.

The platform is designed as a **working proof of concept (PoC)** that can evolve into an enterprise automation solution through integration with existing project-management, collaboration, document, security, and business systems.

1. The Business Problem

Project and engineering teams often spend significant time manually:

- Collecting team status updates
- Monitoring sprint progress
- Identifying project risks
- Following up on stale or overdue updates
- Preparing management reports
- Searching across project documentation
- Reviewing requirements and design documents
- Identifying complexity and potential delivery concerns
- Triggering repetitive project-management workflows

OpenClaw demonstrates an alternative approach:

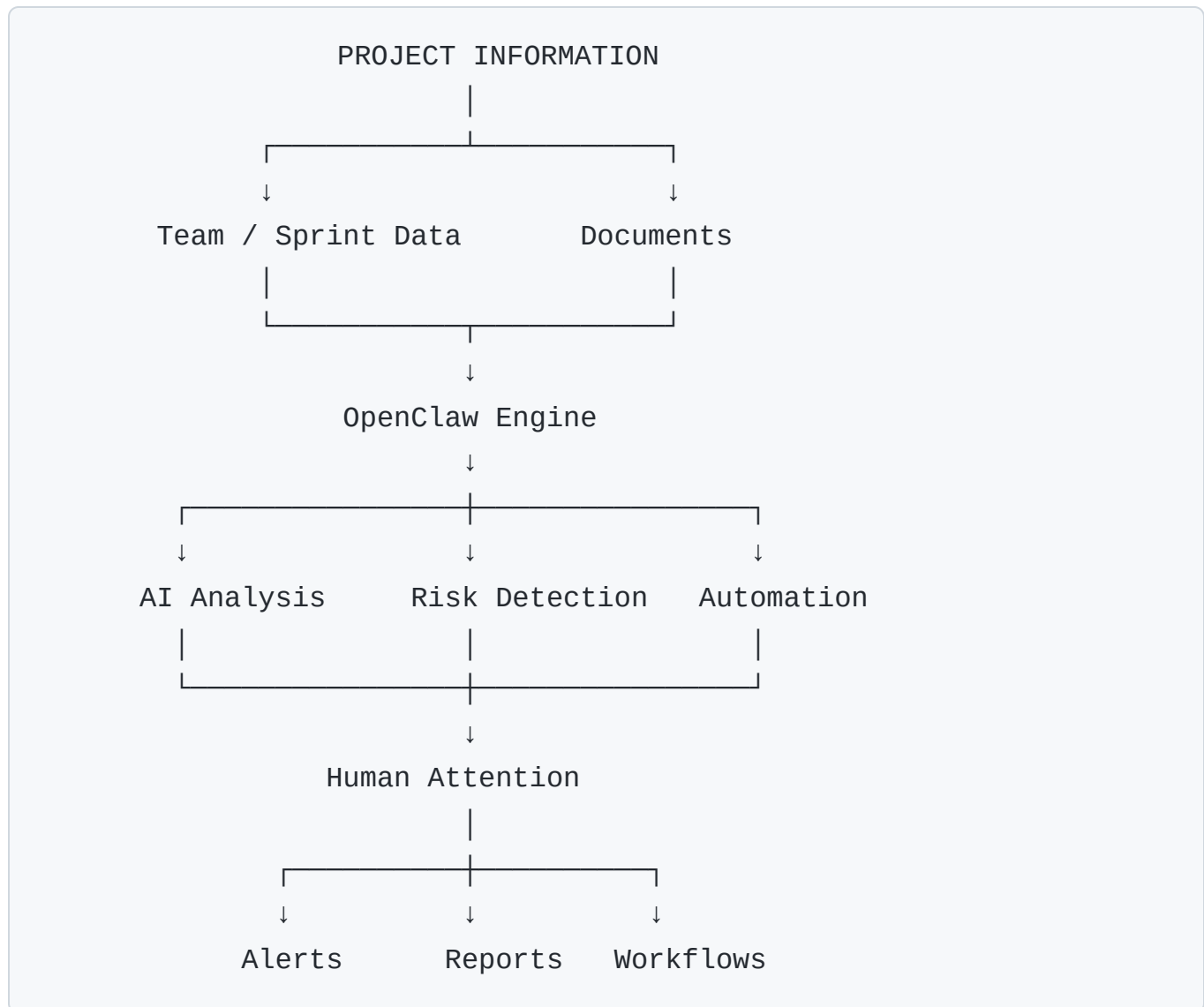
Continuously monitor available project information, use AI to analyse it, identify items requiring attention, and automate appropriate follow-up workflows.

The goal is not to remove the project manager from the process.

The goal is to allow project managers to spend less time collecting and processing information and more time making decisions.

2. What OpenClaw Demonstrates

OpenClaw combines four capabilities:



The current implementation demonstrates:

- AI-assisted project analysis
- Continuous background monitoring
- Risk classification

- Sprint monitoring
- Automated reminders
- Automated daily team digests
- Workflow triggers
- Semantic document search
- AI document summarisation
- Document comparison and synthesis
- Requirements/design complexity analysis
- Role-based access to project information

3. Key Capabilities

Capability	Description
Dashboard	Live team status, risk flags, and activity information
Team Summary	AI-generated team-wide narrative using Mistral
Sprint Calendar	Sprint board with task tracking and velocity statistics
Documents	Version-controlled PDF, DOCX, XLSX, PPTX and TXT document store
Search & Summarise	Semantic search across project documents with AI summarisation and comparison
Complexity Analyser	AI-powered section-level complexity analysis for requirements and design documents
Engine Control	Manager view of autonomous background workers and worker logs
Notes	Personal Markdown notes
File Manager	General file upload/download capability
Risk Classification	AI-assisted classification of team/member risk
Automated Reminders	Detects stale updates and approaching sprint deadlines
Daily Digest	Generates automated team summaries

Capability	Description
Workflow Triggers	Fires automated escalation and sprint-warning events

4. Automation Capabilities

OpenClaw is designed to automate recurring project-operations activities rather than requiring managers to manually monitor every project.

Automated Monitoring

The background engine can:

- Detect stale team status updates
- Identify member/project risk conditions
- Monitor sprint deadline proximity
- Classify risk levels using AI
- Continuously process scheduled checks

Automated Communication

The platform can:

- Generate daily team digests
- Send reminders for stale updates
- Surface sprint warnings
- Trigger escalation workflows

Automated Document Intelligence

The platform can:

- Automatically index documents
- Extract text from supported document formats
- Generate semantic embeddings
- Search project knowledge semantically
- Summarise documents
- Compare documents

- Synthesise information
- Analyse requirements and design complexity

Automated Workflow Triggers

The system includes background workflow workers for:

- Escalations
- Sprint warnings
- Scheduled monitoring
- Reminder generation
- Digest generation

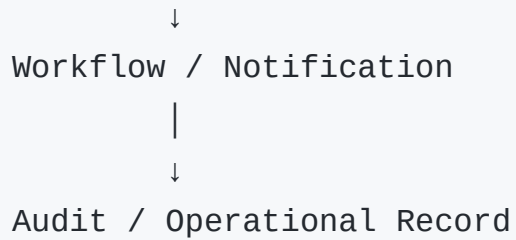
The objective is to move project operations from **manual monitoring toward continuous AI-assisted automation**.

5. Human-in-the-Loop Automation

OpenClaw is designed as an **AI-assisted automation platform**, rather than an uncontrolled autonomous decision-making system.

A typical workflow can follow this pattern:





For a production implementation, workflows can be configured so that low-risk repetitive activities are automated while consequential actions require human approval.

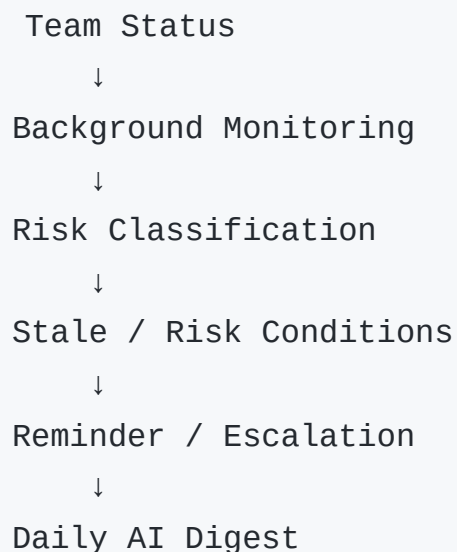
6. Example Business Scenario

Consider an engineering programme with multiple team members and active sprints.

A project manager normally needs to:

1. Collect status updates
2. Review blockers
3. Identify risks
4. Check overdue tasks
5. Review sprint deadlines
6. Search project documentation
7. Prepare a management summary
8. Follow up with individual team members

OpenClaw can automate portions of this workflow:





Manager Review

At the same time:

Requirements / Design Documents



Document Indexing



Semantic Search



AI Analysis



Complexity / Risk Indicators



Manager Review

This provides a demonstration of how AI can support both **project operations and project knowledge management**.

7. High-Level Architecture

Browser

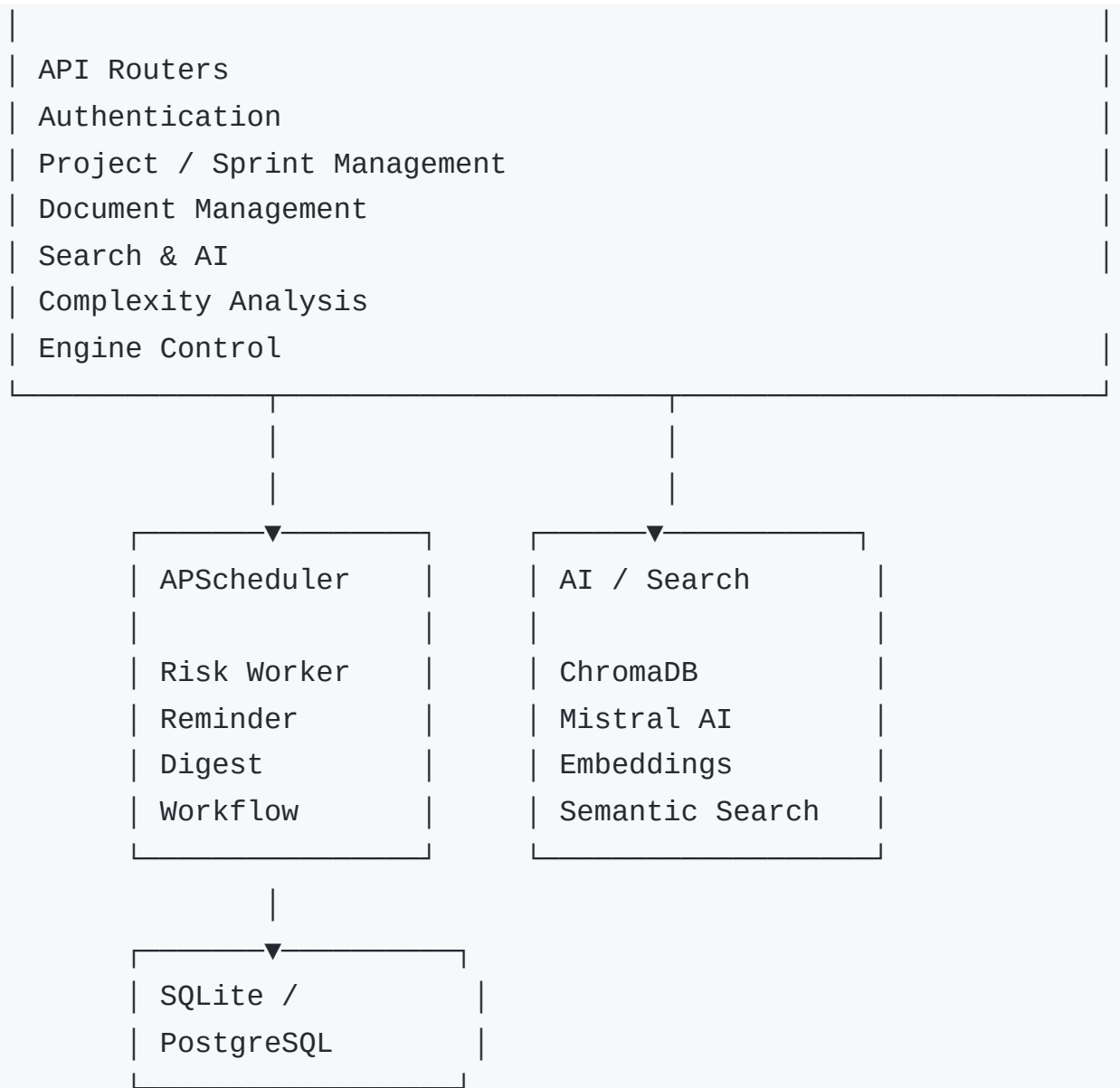
React + Vite

Dashboard · Summary · Calendar · Documents · Search

Complexity · Engine · Notes · Files

HTTP / JSON

FastAPI

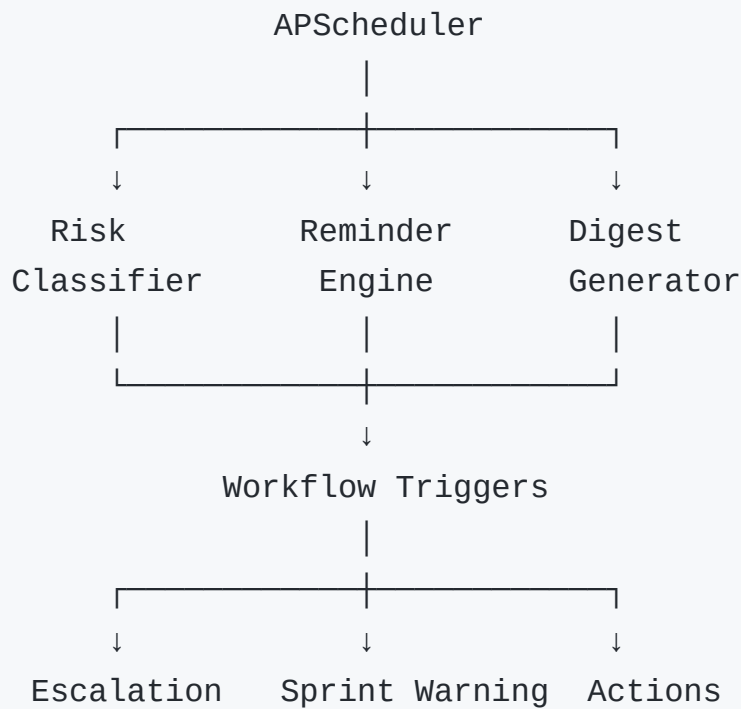


8. Request Flow

1. React SPA authenticates using JWT.
2. Subsequent API requests include a bearer token in the Authorization header.
3. FastAPI validates the authentication token.
4. Requests are dispatched to the appropriate API router.
5. AI/search operations use Mistral and ChromaDB.
6. Background workers run independently of HTTP requests.
7. Scheduled workers evaluate project conditions and trigger configured workflows.

9. Automation Engine

The background engine is one of the core components of OpenClaw.



Current workers include:

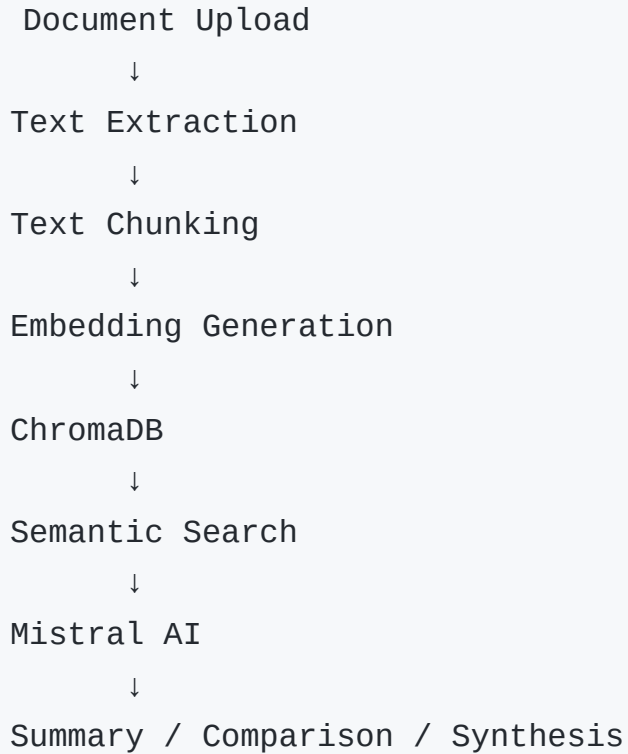
- Risk classifier
- Reminder engine
- Daily digest generator
- Workflow trigger engine

10. Document Intelligence Pipeline

Supported document types include:

- PDF
- DOCX
- XLSX
- PPTX
- TXT

The processing pipeline is:



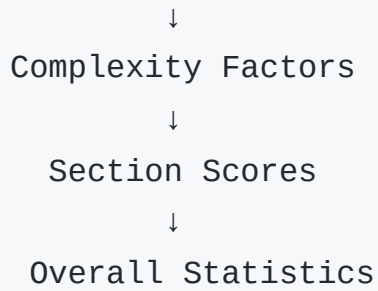
This allows project teams to search and analyse information across multiple project documents rather than manually reviewing each file.

11. Complexity Analysis

OpenClaw includes an AI-powered complexity analyser for requirements and design documents.

The processing flow is:





The intention is to help project teams identify areas of documentation that may require additional review or clarification.

The resulting analysis should be treated as **decision support**, not as an automated replacement for engineering judgement.

12. Backend Structure

```
backend/  
├─ main.py  
├─ config.py  
│  
├─ api/  
│   ├─ auth.py  
│   ├─ status.py  
│   ├─ sprint.py  
│   ├─ documents.py  
│   ├─ search.py  
│   ├─ complexity.py  
│   ├─ engine.py  
│   └─ notes_files.py  
│  
├─ db/  
│   ├─ models.py  
│   ├─ document_models.py  
│   ├─ session.py  
│   └─ seed.py  
│
```

```
├─ engine/
│   ├── scheduler.py
│   └── workers/
│       ├── risk_classifier.py
│       ├── reminder_engine.py
│       ├── digest_generator.py
│       └── workflow_triggers.py
└─ search/
    ├── extractor.py
    ├── chunker.py
    ├── chroma_store.py
    ├── pipeline.py
    ├── startup_indexer.py
    ├── summariser.py
    ├── complexity_analyser.py
    └── complexity_models.py
```

13. Frontend Structure

```
frontend/src/
├─ App.jsx
├─ AuthContext.jsx
├─
├─ components/
│   └── Sidebar.jsx
├─
└─ pages/
    ├── Dashboard.jsx
    ├── TeamSummary.jsx
    ├── CalendarPage.jsx
    ├── DocumentsPage.jsx
    ├── SearchPage.jsx
    └── ComplexityPage.jsx
```

```
|— EngineControl.jsx
|— NotesPage.jsx
└— FilesPage.jsx
```

14. Technology Stack

Layer	Technology
Backend	FastAPI
Server	Uvicorn
ORM	SQLAlchemy 2.0 async
Development Database	SQLite +aiosqlite
Production Database Option	PostgreSQL + asyncpg
Vector Store	ChromaDB
AI	Mistral AI
Embeddings	Mistral Embed
Background Jobs	APScheduler
Authentication	JWT
Password Hashing	bcrypt
Frontend	React 18 + Vite
Document Parsing	PyMuPDF, python-docx, openpyxl, python-pptx
Search	ChromaDB + semantic embeddings
API	REST / HTTP JSON

15. Security

The current prototype includes:

- JWT-based authentication
- Bearer-token API authentication
- Password hashing using bcrypt
- Role separation between managers and developers
- Environment-based configuration
- Configurable secret keys
- Fernet-based token encryption capability

Security configuration includes JWT and encryption settings in the environment configuration.

Production Security Evolution

A production implementation should additionally consider:

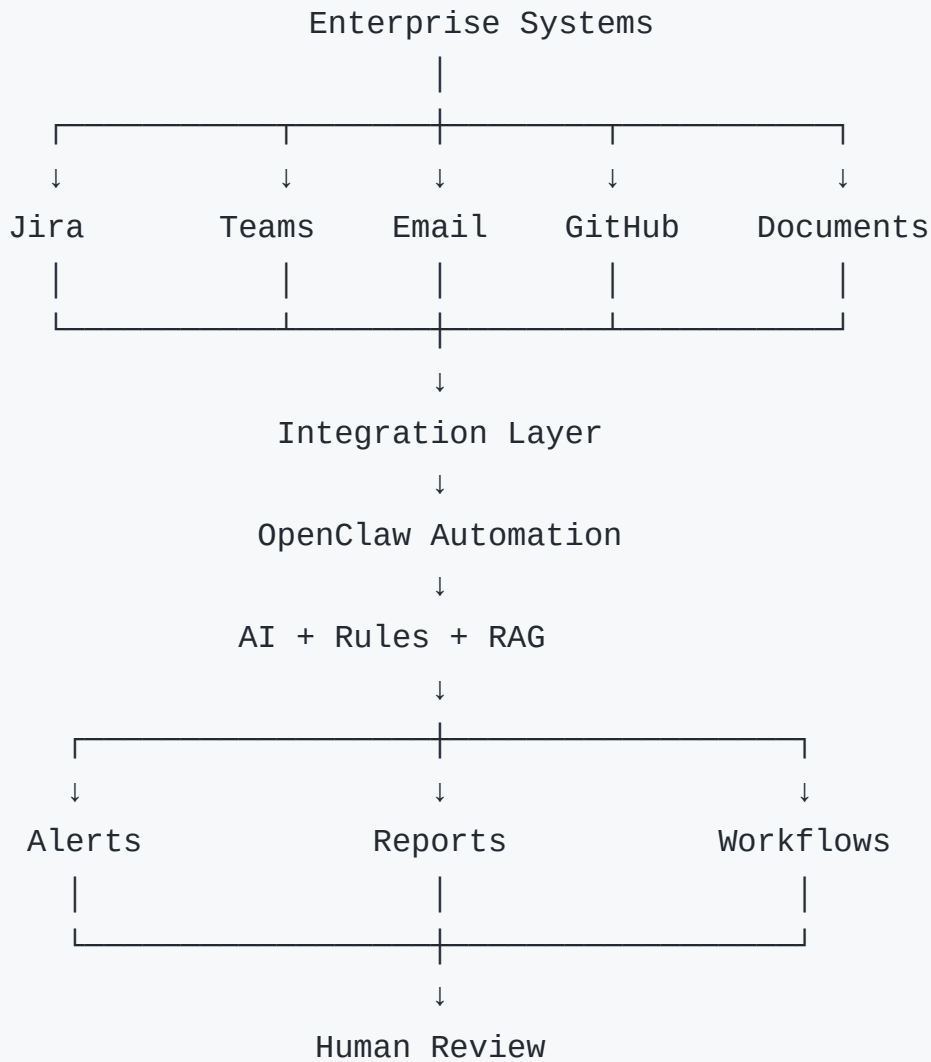
- HTTPS / TLS
- Enterprise identity provider / SSO
- Secure secret management
- Database encryption
- Audit logging
- Rate limiting
- Security headers
- Dependency vulnerability scanning
- Centralized application logging
- Backup and recovery
- Least-privilege access
- Production-grade file/object storage
- Network segmentation where appropriate

The prototype should therefore be viewed as a **working PoC rather than a production security baseline**.

16. Enterprise Integration Path

The current implementation is intentionally lightweight and self-contained.

A future enterprise implementation could integrate OpenClaw with existing systems such as:



These integrations are part of the **potential production evolution** of the platform and are not claimed as current OpenClaw functionality unless implemented separately.

17. Production Evolution

The current project is designed as a working PoC.

Area	Current PoC	Production Evolution
Database	SQLite	PostgreSQL / enterprise database
Vector Store	Local ChromaDB	Managed or enterprise vector infrastructure

Area	Current PoC	Production Evolution
AI	Mistral API	Enterprise AI strategy and model governance
Authentication	JWT	Enterprise SSO / identity provider
Background Jobs	APScheduler	Distributed workflow/job platform
File Storage	Local storage	Object / enterprise document storage
Deployment	Local development	Cloud/containerised deployment
Monitoring	Worker/application logs	Centralised observability
Integrations	Internal APIs	Enterprise systems and APIs
Security	Prototype controls	Full enterprise security architecture

The objective is to demonstrate the **automation and AI workflow first**, while maintaining a clear path toward production architecture.

18. Automation KPIs

A production implementation should measure business impact rather than simply measuring AI activity.

Potential KPIs include:

- Reduction in manual status-report preparation time
- Reduction in time required to identify project risks
- Reduction in overdue project actions
- Time saved searching project documentation
- Reduction in repetitive project-management activities
- Percentage of project updates processed automatically
- Number of automated workflow events
- AI recommendation acceptance rate
- Number of issues surfaced before manual escalation
- Manager time saved per reporting cycle

Actual improvements should be established through a baseline measurement during a client implementation.

19. Installation & Setup

Prerequisites

- Python 3.12
- Node.js 18+
- Mistral AI API key

1 — Clone and Configure

```
git clone https://github.com/denisgerad/openclaw-team.git
cd openclaw-team
```

Copy the example environment file:

```
cp backend/.env.example backend/.env
```

Edit `backend/.env` with the required configuration.

20. Backend

```
cd /path/to/openclaw-team

python3.12 -m venv backend/.venv

source backend/.venv/bin/activate

pip install -r backend/requirements.txt
```

Seed the development database:

```
python -m backend.db.seed
```

21. Frontend

```
cd frontend  
npm install
```

22. Running in Development

Open two terminals.

Terminal 1 — Backend

```
cd /path/to/openclaw-team  
  
source backend/.venv/bin/activate  
  
uvicorn backend.main:app --reload --port 8000
```

Terminal 2 — Frontend

```
cd frontend  
  
npm run dev
```

Open:

```
http://localhost:3000
```

Database tables are created automatically during backend startup.

23. Network Server

To make the backend available to other machines:

```
uvicorn backend.main:app \  
  --reload \  
  --host 0.0.0.0 \  
  --port 8000
```

Configure the frontend development server:

```
server: {  
  host: "0.0.0.0",  
  port: 3000,  
  proxy: {  
    "/api": "http://localhost:8000"  
  }  
}
```

Run:

```
npm run dev -- --host 0.0.0.0
```

Find the server IP:

```
ip addr show eth0
```

Other machines can then connect to:

```
http://<server-ip>:3000
```

24. WSL on Windows

If running the application inside WSL, Windows port forwarding can be configured from an elevated PowerShell session.

Replace `<WSL_IP>` with the WSL interface address:

```
netsh interface portproxy add v4tov4 `
    listenport=3000 `
    listenaddress=0.0.0.0 `
    connectport=3000 `
    connectaddress=<WSL_IP>
```

```
netsh interface portproxy add v4tov4 `
    listenport=8000 `
    listenaddress=0.0.0.0 `
    connectport=8000 `
    connectaddress=<WSL_IP>
```

To remove the forwarding:

```
netsh interface portproxy delete v4tov4 `
    listenport=3000 `
    listenaddress=0.0.0.0
```

```
netsh interface portproxy delete v4tov4 `
    listenport=8000 `
    listenaddress=0.0.0.0
```

25. Demo Accounts

Demo accounts are created by the database seed script.

For security reasons, passwords should not be published in a public repository.

Use the local development seed configuration to create or reset demo users.

Managers have access to Engine Control and can view team-level information.

26. Environment Variables

Copy:

```
backend/.env.example
```

to:

```
backend/.env
```

Example configuration:

```
# Mistral AI
MISTRAL_API_KEY=your-mistral-api-key-here
MISTRAL_MODEL=mistral-small-latest

# Database
DATABASE_URL=sqlite+aiosqlite:///./openclaw.db

# Production database example
# DATABASE_URL=postgresql+asyncpg://user:password@localhost:5432/openclaw

# Security
SECRET_KEY=change-this-to-a-random-64-char-hex-string
```

```
JWT_ALGORITHM=HS256
JWT_EXPIRE_MINUTES=480

TOKEN_ENCRYPTION_KEY=your - fernet - key - here

# Application
APP_NAME=OpenClaw
APP_ENV=development

# Daily digest recipients
DIGEST_RECIPIENTS=manager@yourcompany.com

# Optional web search
WEB_SEARCH_ENABLED=true

# Optional Google OAuth
# GOOGLE_CLIENT_ID=
# GOOGLE_CLIENT_SECRET=
# GOOGLE_REDIRECT_URI=http://localhost:8000/api/auth/oauth/callback
```

Never commit real API keys, passwords, encryption keys, or production secrets to source control.

27. Suggested Client Demonstration

For an enterprise consulting audience, the recommended demonstration flow is:

1. Start with the business problem

Explain the amount of manual effort required to monitor projects, identify risks, prepare reports, and search project documentation.



2. Show the dashboard

Demonstrate:

- Team status
- Risk indicators
- Project activity
- Sprint information

3. Introduce the automation engine

Show that the system can operate in the background rather than requiring the manager to manually initiate every analysis.

4. Demonstrate a risk scenario

Show how project/member information can result in a risk classification or workflow event.

5. Demonstrate document intelligence

Upload or select a requirements/design document and demonstrate:

- Semantic search
- AI summary
- Comparison
- Complexity analysis

6. Generate the team digest

Demonstrate how information collected from the system can be converted into an executive-friendly summary.

7. Show the enterprise evolution

Explain how the same architecture could eventually connect to:

- Project-management platforms
- Collaboration tools
- Enterprise document repositories
- Business systems
- Security platforms

8. Close with measurable outcomes

Discuss how a production deployment would establish baseline measurements and track:

- Time saved
- Risk identification
- Reporting effort
- Workflow automation
- Manager productivity

28. Project Positioning

OpenClaw should be viewed as a **technical demonstration of an AI-powered project-operations automation approach**, rather than as a replacement for existing enterprise project-management platforms.

Its purpose is to demonstrate how an organisation can introduce an intelligent automation layer around existing processes and systems.

The underlying concept is:

Existing Enterprise Systems



Data / Events



AI + Automation Layer



Analysis / Decisions



Human Review Where Needed



Automated Action



Measurement / Audit

This architecture can be adapted to different business processes beyond project management.

29. Potential Extension Areas

The same architecture can be extended to other enterprise automation use cases, including:

Project Operations

- Project health monitoring
- Risk identification
- Automated reporting
- Action tracking

Document Intelligence

- Contract analysis
- Requirements analysis
- Policy analysis
- Knowledge retrieval

Business Process Automation

- Event-driven workflows
- Notifications
- Approvals
- Escalations
- Scheduled processing

Cybersecurity

- Security-event analysis
- Risk classification
- Alert enrichment
- AI-assisted investigation

These are potential extensions and should be implemented and validated separately before being presented as production capabilities.

30. Project Status

Current status: Working Proof of Concept

The current implementation demonstrates:

- Full-stack web application
- REST API architecture
- Authentication
- Project/sprint management
- AI-assisted analysis
- Document intelligence
- Semantic search
- Complexity analysis
- Background automation
- Risk classification
- Automated reminders
- Automated digests
- Workflow triggers

The next stage for enterprise deployment would be integration, security hardening, observability, scalability, governance, and measurable business-value validation.

31. Vision

OpenClaw demonstrates a simple principle:

AI should not only answer questions. It should continuously help organisations understand what is happening, identify what requires attention, and automate appropriate actions.

The long-term direction is an **AI-assisted enterprise automation layer** that works alongside existing systems while keeping people responsible for important decisions.